From Regional Dialects to Standard Written Marathi: A Normalization pipeline based on sequence-to-sequence fine-tuning

Aditya Kinjawadekar[1], Surender Kannaiyan[1], and Saugata Sinha[1]

[1]Visvesvaraya National Institute of Technology, Nagpur, India

E-mail: kinjawadekaradi112@gmail.com

Abstract

Scarcity of resources and absence of benchmarks for regional dialects of Indic languages is an established problem. In this work, we present an end-to-end neural fine-tuning pipeline for normalization of dialectal transcripts from the Marathi dialects of Southern Konkan, Northern Konkan, and Varhadi into Standard Marathi. To address the issues regarding data, we introduce a synthetic data generation pipeline with a multi-tier verification engine for generating the aligned corpus and expanding it further. We evaluate IndicBART (244M) and mT5-small (300M) across varied configurations. Experimental metrics show that mT5-small achieves 69.51 BLEU and 84.58 chrF++ on the expanded multi-dialect benchmark, outperforming IndicBART by 12.39 BLEU. Also, the multi-tier verification prevents the quality drop (+14.03 BLEU recovery over unverified data), suggesting the importance of verification of data during synthetic augmentation for dialect normalization.

Keywords: Dialect Normalization, Low-Resource NLP, Marathi Dialects, Sequence-to-Sequence Modelling, Synthetic Data Augmentation, mT5, IndicBART.

1 Introduction

Rapid development of deep learning architectures and self-attention mechanisms (Vaswani et al., 2017) has made it possible to develop systems capable of understanding and generating natural language. Open-weight models have gained popularity to process text in various domains as they provide a finer control over system architecture, runtime environment, and training data (Dubey et al., 2024). Yet current language systems tend to work efficiently with standard high-resource languages. Queries based on non-standard input yield unexpected and undesired outputs (Joshi et al., 2020). Speakers of regional dialects face many issues in systems that use these models across automated speech recognition, machine translation and other language processing systems (Koenecke et al., 2020; Zampieri et al., 2020).

Dialect normalization is the task of translating non-standard regional dialects into standard written language. The main challenge to be addressed while designing such systems is the preservation of the meaning of original text (Blaschke et al., 2024). It serves to be an essential layer for digital linguistic inclusion. In countries having large and diverse speech communities, such as India, which has 22 official languages and hundreds of regional dialects spoken by 1.4 billion people, regional varieties have remained unattended in computational linguistic research. An example of such linguistic disparity is Marathi, an Indo-Aryan language spoken by over 83 million people in western India (Kulkarni-Joshi, 2023). Development of reliable pipelines for such regional Indic dialects has faced these fundamental challenges:

- *Resource Scarcity*: Parallel corpora mapping the regional dialects to standard written language do not exist. Curation of such sets by experts fluent in multiple dialects is both expensive and cannot be scaled (Blaschke et al., 2024; Dhasmana et al., 2026).
- *Grammatical Variance*: Marathi exhibits rich morphology where different words undergo varied complex transformations across different regions where the dialects are spoken

(Ansary et al., 2024)(Joshi et al., 2025).

- *Degraded performance of Multilingual Models*: Performance of pre-trained models like IndicBART (Dabre et al., 2022), IndicNLP Suite (Kakwani et al., 2020), and mT5 (Xue et al., 2021) seems to drop drastically on non-standard language variants due to fragmentation during tokenisation and out-of-vocabulary phonological shifts (Joshi et al., 2020; Kumar et al., 2024).

In this work, we investigate the normalization performance of IndicBART and mT5 across regional Marathi dialects and evaluate how efficiently they adapt to dialect-to-standard mapping under task-specific conditioning, and check whether synthetic data can solve severe data scarcity while maintaining considerable performance.

Our core contributions are:

- *Parallel Corpus Generation*: Based on transcripts from the RESPIN dataset (Kumar et al., 2025), we construct a parallel corpus for three of the Marathi dialects (Southern Konkan, Northern Konkan, and Varhadi) mapped to Standard Marathi.
- *Automated Synthetic Augmentation with Verification*: We design a data expansion pipeline paired with a multi-tier verification engine, which has rule-based script filtration and LLM-based semantic inspection.

2 Related Work

2.1 Regional Dialect Standardization across Language Families

Early approaches in the context of European languages used Character-Level Statistical Machine Translation (CSMT) and phrase-based models for Swiss German Dialects (Scherrer and Ljubešić, 2016) which were later replaced by pre-trained language models (Lusetti et al., 2018; Kresic and Abbas, 2024). Similar approaches have been developed for Finnish, Swedish, and Estonian dialects (Partanen et al., 2019; Hämäläinen et al., 2020, 2022). A study of four language families by Kuparinen et al. (Kuparinen et al., 2023) suggested that fine-tuned sequence-to-sequence transformers surpassed the rule-based and SMT baselines. Other than European countries, East Asian and African-Asian varieties have also shown similar patterns. Japanese regional dialects have been processed using phrase-level LSTM modules (Abe et al., 2018), while Mandarin scripts relied on acoustic-phonetic mapping (Zhao and Chodroff, 2022). The Conventional Orthography for Dialectal Arabic (CODA) framework (Habash et al., 2012) in the landscape of Arabic languages gave guidelines for Modern Standard Arabic (MSA) which led to multi-city translation benchmarks (MADAR (Bouamor et al., 2018), NADI (Abdul-Mageed et al., 2023)) and morphology-oriented prompt design for LLMs (Eryani et al., 2020; Dimakis et al., 2025).

2.2 Indic Dialect Landscape and Marathi NLP

Even though pre-trained models such as IndicBART (Dabre et al., 2022) and mT5 (Xue et al., 2021), which are trained on multilingual corpora cover major languages, they lack understanding of regional sub-dialects. Models based on a single language like L3Cube-MahaNLP (Joshi et al., 2022) offer pre-trained Marathi models for classification, but do not support generative text standardization. INDIC-DIALECT (Kumar et al., 2024) considers Hindi varieties (Bhojpuri, Magahi, Maithili), for evaluation via WFST frameworks for text normalization while preserving the underlying semantics (Pratap et al., 2024). However, Marathi dialects were primarily dealt for acoustic/ASR classification (Mulla and Kumar, 2022; Pawar et al., 2026), or zero-shot LLM retrieval (Gaikwad et al., 2025; Vyavhare et al., 2026). Our work fills this gap by presenting an end-to-end neural fine-tuning pipeline with evaluations and data augmentation strategies.

2.3 LLM Distillation and Multi-Tier Verification

LLMs can be used to suffice for data scarcity in various tasks (Wang et al., 2022; Taori et al., 2023; Mukherjee et al., 2023). Knowledge distillation strategies can be employed to train compact and efficient models based on the outputs generated by a larger teacher model (Hsieh et al., 2023; Pecher et al., 2024; de Gibert et al., 2024). But the use of such techniques for low-resource regional dialects involves significant amount of hallucinations and noise (de Gibert et al., 2024; Ji et al., 2023). We solve this by utilising Gemma-4 (Google DeepMind Gemma Team, 2026) with a robust morphologically aware system prompt and a multi-tier verification system that ensures integrity of the data generated.

3 Methodology

3.1 Original Corpus Curation and Foundation in RESPIN

Base sentences for our task are drawn from the domain-specific transcripts of the RESPIN-S1.0 initiative (Kumar et al., 2025), developed at IISc Bangalore under the Gates Foundation project. While RESPIN has audio recordings and their transcripts across agriculture and banking domains for Indian languages and their dialects, it lacks aligned parallel dialect-to-target language translations required for efficient normalisation. The large speaker diversity keeps a sufficiently diverse coverage of intra-dialect variations, while the balanced domain splits reduce systematic bias in our derived parallel corpus. Detailed corpus statistics are summarized in Table 1 and Figure 1.

Table 1: RESPIN-S1.0 Marathi Corpus Summary

Code	**Dialect Name**	**Utterances (%)**	**Duration (h)**	**Speakers**	**Prompts**
D1	Southern Konkan (Malvani)	203,651 (25.14%)	247.31	732	5,838
D2	Northern Konkan (Ahirani)	198,560 (24.52%)	265.25	557	5,825
D3	Standard Marathi (Puneri)	208,850 (25.79%)	255.79	608	6,077
D4	Varhadi (Vidarbha)	198,873 (24.55%)	257.70	747	5,330
Total	**All Dialects**	**809,934 (100%)**	**1,026.06**	**2,644**	**23,069**

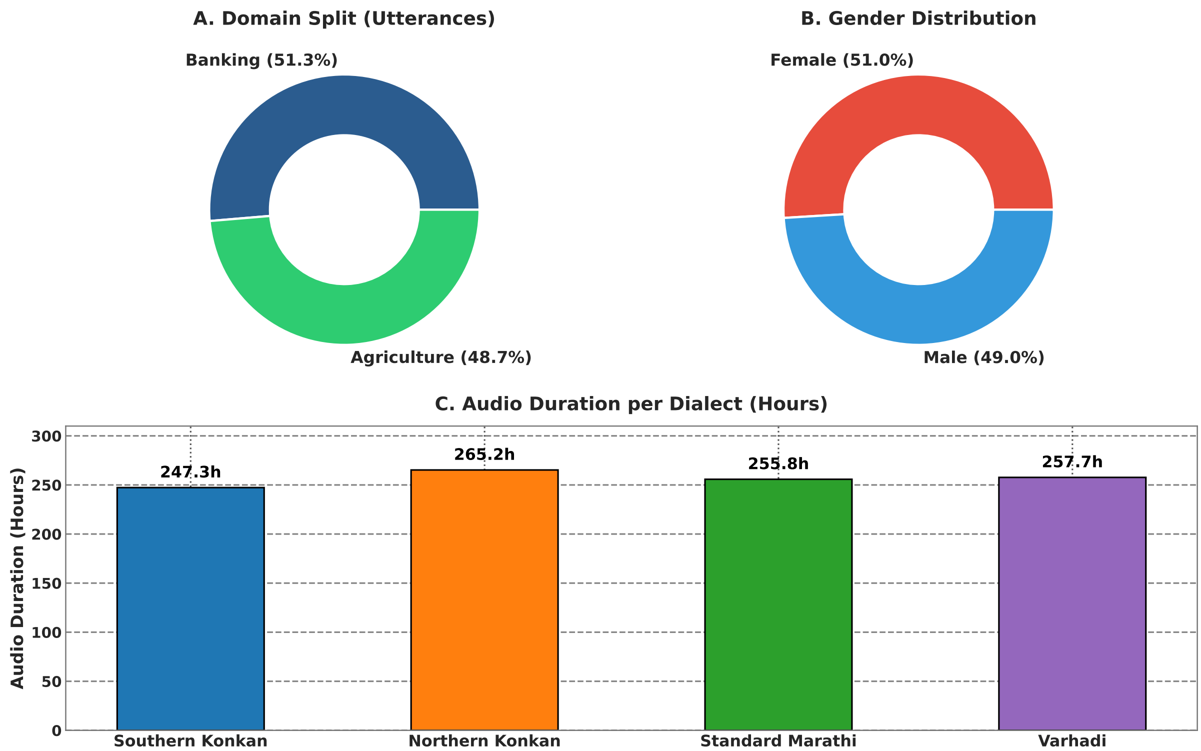

Figure 1: Corpus demographics: (A) domain split, (B) gender distribution, and (C) audio duration per dialect.

3.2 Target Dialects and Linguistic Properties

We target the following Marathi dialects based on the RESPIN corpus:

- *Southern Konkan*: Known for its shortening of final-word vowels, softening of dental consonants, and different auxiliary verb forms. Differences include systematic vowel shifts: standard जातो ("he goes") becomes Southern Konkan जातां; standard कुठे ("where") becomes कुठंय.
- *Northern Konkan*: Heavily influenced by neighbouring Gujarati and Bhili language families, thus it has consonant simplification and dental consonant shifts. Some variations include: standard कशाला ("why") → Northern Konkan कसाले; standard मुलगा ("boy") → Northern Konkan पोर.
- *Varhadi*: Spoken in the regions of Vidarbha, it shows very specific verb suffix modifications. Standard interrogative का? ("why?") becomes Varhadi काऊन?.

3.3 Synthetic Parallel Data Generation Pipeline

To deal with data scarcity, we developed an automated synthetic data translation and generation pipeline powered by Gemma-4 (Google DeepMind Gemma Team, 2026) with 12B parameters configuration, accessed via Ollama. The complete end-to-end normalization architecture is illustrated in Figure 2. We considered several design decisions to ensure generation quality to be maximised:

- *1-Call-Per-Sentence Generation*: We observed batch generation caused hallucinations in context. Therefore, a single API call was utilized for a single sentence, flushing the KV-cache at every instant that gave fresh context to the LLM every time.
- *Preservation of Meaning*: System prompts were designed in a way that actually instructed the model that while the dialectal text is to be translated into standard written Marathi, the standard output must preserve the exact semantic meaning, domain-specific terminology (agriculture/banking), and proper nouns without any change.

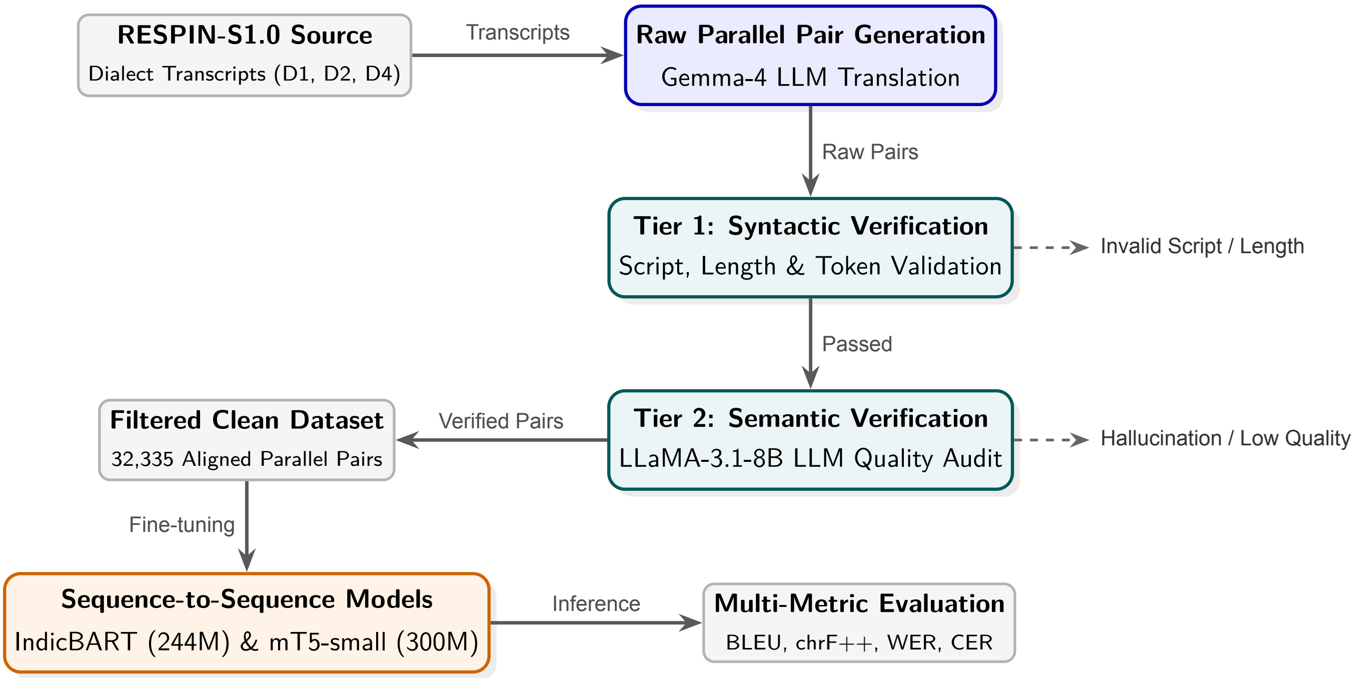

Figure 2: Dialect normalization pipeline: synthetic data generation, two-tier verification, and model training.

3.4 Verification and Quality Filtering Engine

LLMs can generate flawed target sequences when dealing with regional sub-dialects. We implemented a two-tier verification engine to remove them before model training:

- *Tier 1: Rule-Based Syntactic Filter.* Automated validation checks implement Indic Abugida script normalization principles (Ansary et al., 2024), including:
  - *Script Consistency*: Discards samples containing non-Devanagari characters.
  - *Length Ratio Constraint*: Discards pairs where sentence length ratio $\frac{\text{len(Target)}}{\text{len(Source)}} \notin [0.5, 2.0]$.
  - *Minimum Token Count*: Both source and target must contain ≥ 3 whitespace-separated tokens.
- *Tier 2: LLM Semantic Verification.* A secondary LLM evaluator using `llama-3.1-8b-instant` (Dubey et al., 2024) evaluates the accuracy on a scale of 1–5 under redesigned system prompts. Pairs with semantic scores < 3 are discarded.

Through this two-step process, 3,751 low-quality sentence pairs were completely discarded from the corpus (19,914 pairs), preventing further error propagation. The dataset composition across final clean partitions is displayed in Figure 3. The same synthetic generation and verification pipeline was subsequently utilized to expand the parallel dataset from 16,163 original pairs to 32,335 verified clean pairs across all three dialects.

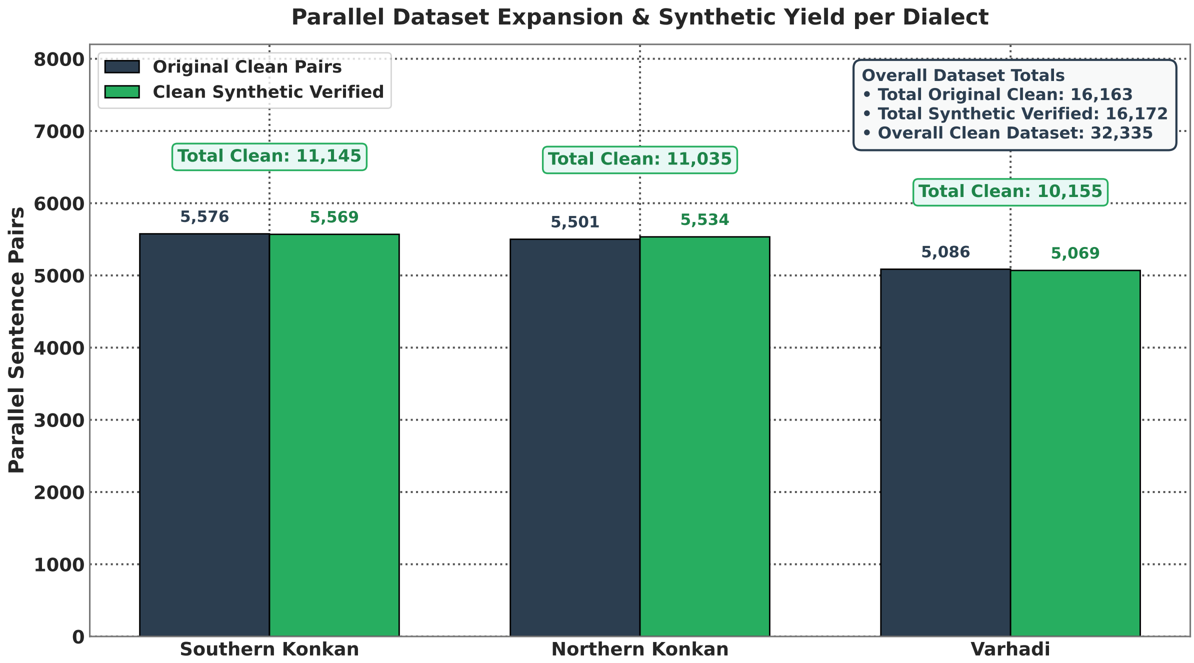

Figure 3: Dataset sizes per dialect before and after synthetic expansion.

3.5 Model Fine-Tuning and Conditioning

We evaluate two pre-trained sequence-to-sequence models across different configurations: IndicBART (244M) (Dabre et al., 2022) and mT5-small (300M) (Xue et al., 2021).

To enable unified multi-dialect training where a single model is used for all regional variants, we design special task-based conditioning prefixes:

- *Task Normalization Prefixes*: A normalisation tag (`<normalize>`) is attached that instructs the encoder to perform the dialect-to-standard text transformation.
- *Dialect-Source Identification Tags*: Source sentences have dialect markers, `<D1>` for Southern Konkan, `<D2>` for Northern Konkan, and `<D4>` for Varhadi for explicit conditioning of the models in multi-dialect training and testing scenarios.
- *Target Standardization Tags*: Decoder target is conditioned using standard language tags (`<standard_mr>`), enforcing the output alignment into Standard Written Marathi.

Table 2 considers the underlying properties based on the architecture and hyperparameter configurations for the fine-tuning of IndicBART (244M) and mT5-small (300M).

Table 2: Architecture comparison: IndicBART (244M) vs. mT5-small (300M)

Property	**IndicBART (244M)**	**mT5-small (300M)**
Base Architecture	mBART-50 (BART Enc-Dec)	T5 Encoder-Decoder
Position Encoding	Absolute sinusoidal	Relative positional bias
Vocabulary Size	64,000 SentencePiece	250,000 SentencePiece (mC4)
Pre-training Data	11 Indic + English	101 languages (mC4)
Script Approach	Devanagari-unified	Native multi-script
Target Conditioning	Required `<2mr>` token	Raw text-to-text (no prefix)
Learning Rate	5×10^{-5} (linear warmup)	5×10^{-4} (AdaFactor)

There is a fundamental trade-off present in the models. IndicBART is tailored for Indic language scripts with a unified 64k Devanagari vocabulary and language-specific target tokens (`<2mr>`), providing stronger performance in single-dialect tasks. However, mT5-small features a substantially larger 250k multilingual vocabulary which reduces subword fragmentation on rare dialectal forms, and relative positional embeddings capture local suffix and verbal shifts efficiently. These inherent structural advantages allow mT5-small to scale its performance in multi-dialect scenarios.

3.6 Evaluation Protocol

We report the model performance across the following two evaluation settings:

- *Mean 5-Fold Cross-Validation*: To evaluate the stability of models during training, they are trained and evaluated under an 85/15 stratified 5-fold cross-validation scheme.
- *Held-Out RESPIN Test Benchmark*: To consider the performance on unseen data, models were trained on the complete dataset and later evaluated on the official held-out set (`IISc_RESPIN_test_mr`), consisting of 2,170 unseen utterances across all target sub-dialects.

Results are reported using metrics that are evaluated using SacreBLEU (Post, 2018) and Word Error Rate (WER) scripts:

- *BLEU* (Bilingual Evaluation Understudy): BLEU-4 with 4-gram precision, brevity penalty, with an exponential smoothing method.
- *chrF++*: Character n-gram F-score combined with word unigrams and bigrams.
- *Normalized WER / CER*: Word Error Rate and Character Error Rate after Devanagari Unicode script standardization.

4 Findings

4.1 Single-Dialect vs. Multi-Dialect 5-Fold Cross-Validation Results

Cross-validation performance on original single-dialect datasets is reported in Table 3.

Table 3: Single-dialect model performance on original datasets (5-Fold CV)

	IndicBART (244M)		mT5-Small (300M)	
Dialect Name	**BLEU**	**chrF++**	**BLEU**	**chrF++**
Southern Konkan (D1)	**48.54**	**78.35**	46.31	74.43
Northern Konkan (D2)	**60.58**	**80.30**	60.31	77.34
Varhadi (D4)	76.63	90.93	**81.00**	**91.21**

Baseline performance based on Table 3 highlights the structural differences in Marathi dialects. Both models achieve their highest scores on Varhadi (76.63–81.00 BLEU and 90.93–91.21 chrF++), whereas Southern and Northern Konkan have shown lower baseline scores (46.31–60.58 BLEU). This indicates the higher order of overlap between Varhadi dialect with standard Marathi when it comes to grammar and lexicography. Table 4 presents the dialectwise breakdown of metrics when training on multi-dialect data.

Table 4: Multi-dialect model performance on original datasets (5-Fold CV)

	IndicBART (244M)		mT5-Small (300M)	
Evaluated Subset	**BLEU**	**chrF++**	**BLEU**	**chrF++**
Southern Konkan (D1)	26.21	53.15	**48.28**	**75.04**
Northern Konkan (D2)	47.59	67.16	**62.35**	**78.92**
Varhadi (D4)	72.75	85.63	**79.57**	**90.51**

Combined training showed a drop in IndicBART metrics, from 48.54 to 26.21 BLEU and 53.15 chrF++ on Southern Konkan, possibly due to cross-dialectal interference. But mT5-small retained performance, maintaining 48.28 BLEU on Southern Konkan and achieving an overall average score of 63.29 BLEU and 81.37 chrF++. Table 5 evaluates single-dialect models trained on synthetically expanded corpora.

Table 5: Single-dialect model performance on expanded datasets (5-Fold CV)

	IndicBART (244M)		mT5-Small (300M)	
Dialect Name	**BLEU**	**chrF++**	**BLEU**	**chrF++**
Southern Konkan (D1)	47.25	64.69	**65.10**	**82.88**
Northern Konkan (D2)	40.51	62.21	**62.07**	**79.29**
Varhadi (D4)	73.62	86.75	**78.89**	**90.59**

IndicBART experiences slight saturation of performance across single-dialect configurations with a score of 47.25 BLEU on Southern Konkan and 40.51 BLEU on Northern Konkan, while mT5-small has utilised the synthetic data more effectively, spiking the scores to 65.10 BLEU (+18.79) on Southern Konkan and 62.07 BLEU on Northern Konkan with chrF++ exceeding 79.29–90.59. Table 6 presents the evaluation of the multi-dialect model trained on the synthetically expanded corpus.

Table 6: Multi-dialect model performance on expanded datasets (5-Fold CV)

	IndicBART (244M)		mT5-Small (300M)	
Evaluated Subset	**BLEU**	**chrF++**	**BLEU**	**chrF++**
Southern Konkan (D1)	52.06	72.15	**66.62**	**83.57**
Northern Konkan (D2)	49.08	69.90	**62.72**	**79.63**
Varhadi (D4)	70.27	84.27	**78.73**	**90.46**

IndicBART recovers performance from the cross-dialect interference to score 52.06 BLEU (+25.85 gain over original multi-dialect) on Southern Konkan and 57.12 overall BLEU, while mT5-small reaches peak performance across all dialects (66.62 BLEU on Southern Konkan, 62.72 BLEU on Northern Konkan, and an overall average of 69.51 BLEU and 84.58 chrF++). Comparative single-dialect 5-fold cross-validation performance shifts are illustrated in Figure 4.

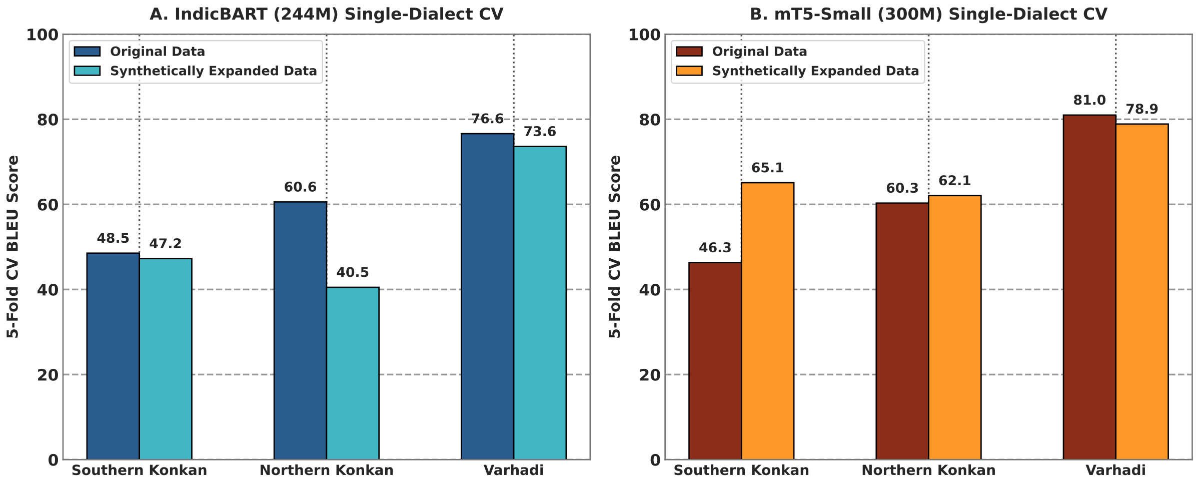

Figure 4: Single-dialect cross-validation scores before and after synthetic expansion.

The multi-dialect performance matrix is shown in Figure 5.

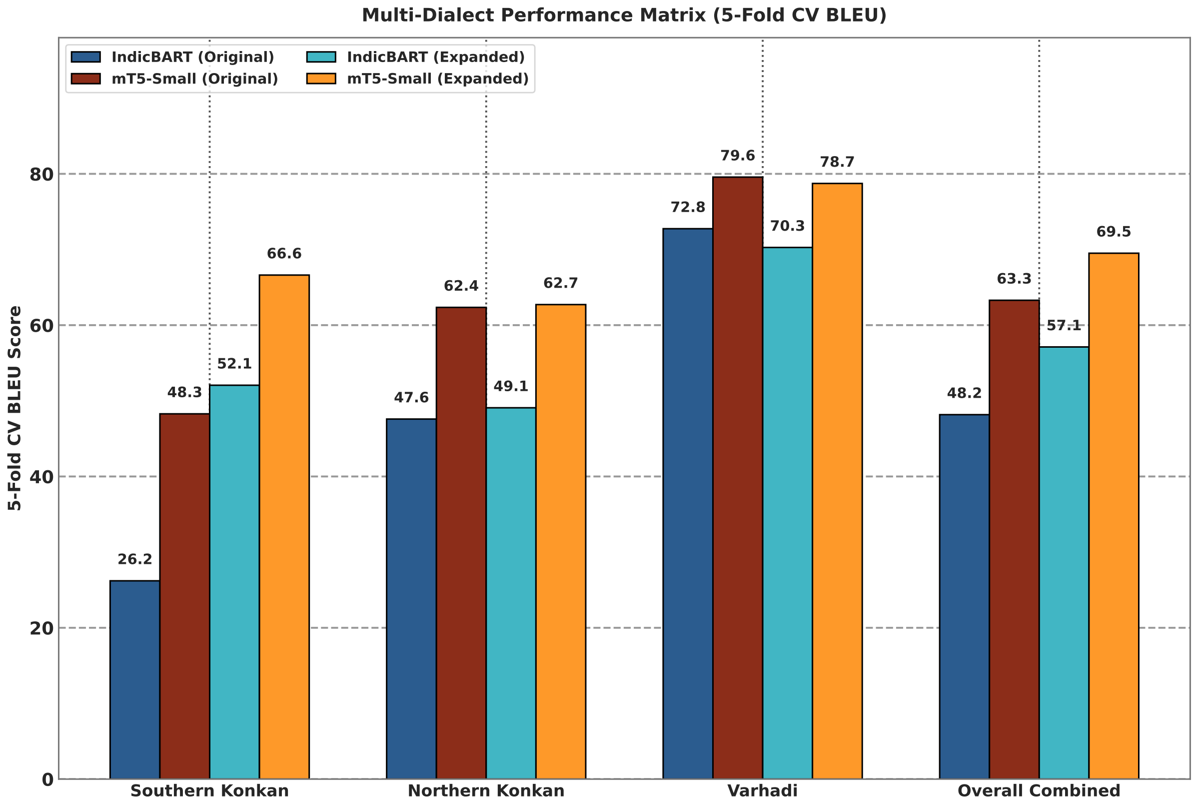

Figure 5: Multi-dialect cross-validation performance across dialects and models.

4.2 Comprehensive Benchmark on the Held-Out Test Set

Subsequent evaluations will be conducted on the official held-out set only after training on the complete datasets without any cross-validation. Single-dialect training shows that synthetic data augmentation substantially reduces IndicBART performance across all three dialects, while mT5-small remains comparatively stable and shows slight improvements on Northern Konkan and Varhadi. Multi-dialect training produces substantial gains for IndicBART on Southern Konkan (+26.0 BLEU) and Northern Konkan (+6.5 BLEU), while Varhadi decreases slightly (−3.0 BLEU). mT5-small shows smaller but consistently positive gains across all three dialects: +3.0, +0.7, and +1.2 BLEU for Southern Konkan, Northern Konkan, and Varhadi, respectively. IndicBART has a degraded performance when augmented with synthetic data while mT5-small remains stable with slight improvements. The full performance evaluation on the held-out test set is shown in Table 7 and Figure 6.

Table 7: Model performance on the held-out RESPIN test set

Model Scope	Training Data	Dialect / Subset	IndicBART (244M)		mT5-Small (300M)	
			BLEU	WER	BLEU	WER
Single	Original	Southern Konkan	**57.80**	**26.60%**	43.86	34.37%
		Northern Konkan	**90.13**	**6.46%**	79.46	11.95%
		Varhadi	**83.59**	**10.19%**	74.81	14.73%
	Expanded	Southern Konkan	24.06	60.32%	**44.36**	**32.75%**
		Northern Konkan	65.41	28.70%	**79.76**	**12.61%**
		Varhadi	67.97	25.43%	**76.90**	**14.19%**
Multi	Original	Southern Konkan	26.08	52.98%	**40.94**	**35.34%**
		Northern Konkan	47.92	32.64%	**77.50**	**14.65%**
		Varhadi	73.04	26.96%	**73.47**	**16.16%**
	Expanded	Southern Konkan	**52.15**	36.50%	43.88	**34.96%**
		Northern Konkan	54.35	25.65%	**78.16**	**13.55%**
		Varhadi	69.96	21.04%	**74.74**	**15.57%**

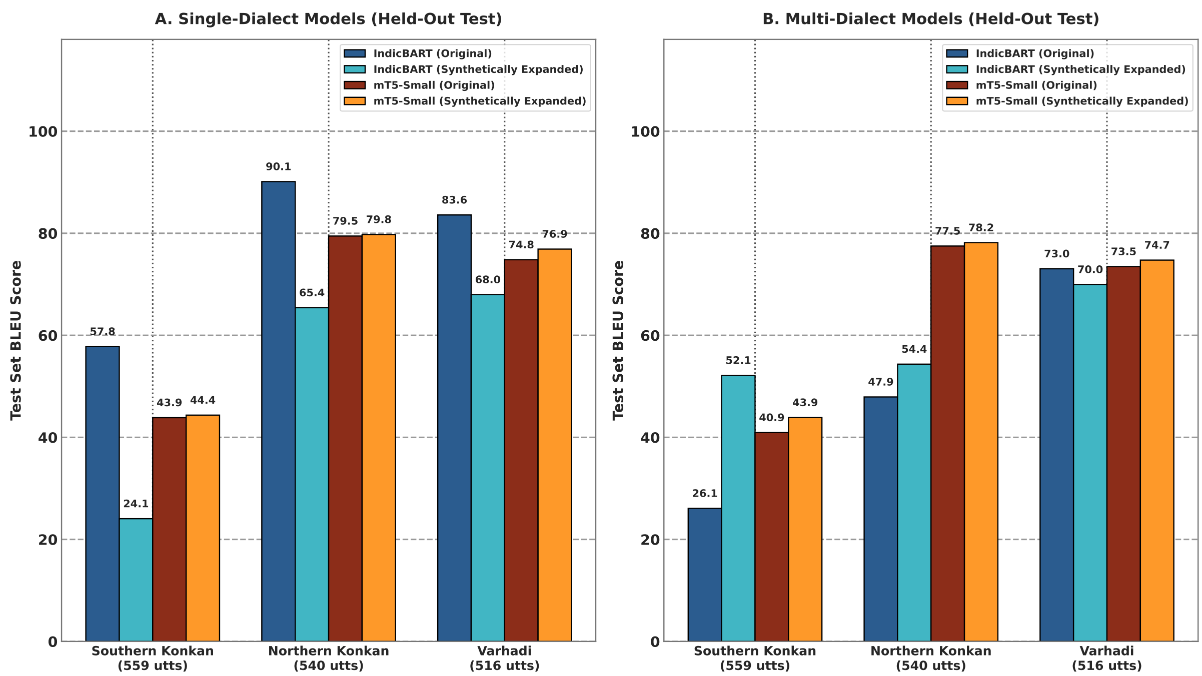

Figure 6: Held-out test set performance across training setups.

4.3 Impact of Synthetic Data Augmentation on Test Set

Table 8 and Figure 7 show the overall impact on metrics by synthetic data augmentation directly on the held-out test set under multi-dialect training schemes. IndicBART showed improvement for Southern Konkan (+26.07 BLEU) and Northern Konkan (+6.43 BLEU), but its Varhadi score dropped by 3.08 BLEU. mT5-small showed consistent improvements: +2.94 BLEU for Southern Konkan, +0.66 BLEU for Northern Konkan, and +1.27 BLEU for Varhadi.

Table 8: Effect of data augmentation on held-out test BLEU

Evaluated Subset	IndicBART BLEU			mT5-Small BLEU		
	Original	Expanded	Δ	Original	Expanded	Δ
Southern Konkan (D1)	26.08	52.15	**+26.07**	40.94	**43.88**	**+2.94**
Northern Konkan (D2)	47.92	54.35	**+6.43**	77.50	**78.16**	**+0.66**
Varhadi (D4)	**73.04**	69.96	−3.08	73.47	**74.74**	**+1.27**

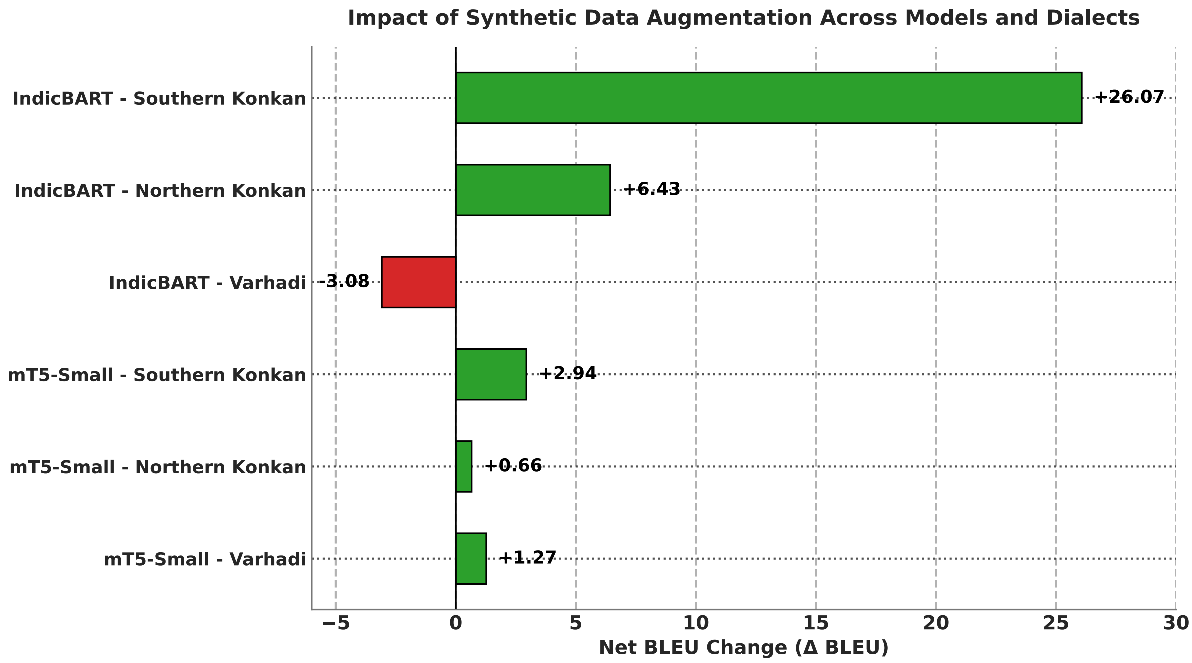

Figure 7: BLEU score changes on the held-out test set after data augmentation.

4.4 Verification Engine Impact: Raw Unverified vs. Filtered Clean Data

We also analyzed the performance of our verification and filtering engine on the test set by training on the contaminated versus the clean dataset. Table 9 and Figure 8 show the BLEU and chrF++ score differences due to multi-tier verification.

Table 9: Impact of data verification on normalization scores

Model	Pipeline State	Pairs	BLEU	chrF++	Δ BLEU	Δ chrF++
IndicBART (244M)	Raw Unverified Data	19,914	43.09	64.54	−14.03	−10.95
	Filtered Clean Data	32,335	**57.12**	**75.49**	**+14.03**	**+10.95**
mT5-Small (300M)	Raw Unverified Data	19,914	61.21	80.18	−8.30	−4.40
	Filtered Clean Data	32,335	**69.51**	**84.58**	**+8.30**	**+4.40**

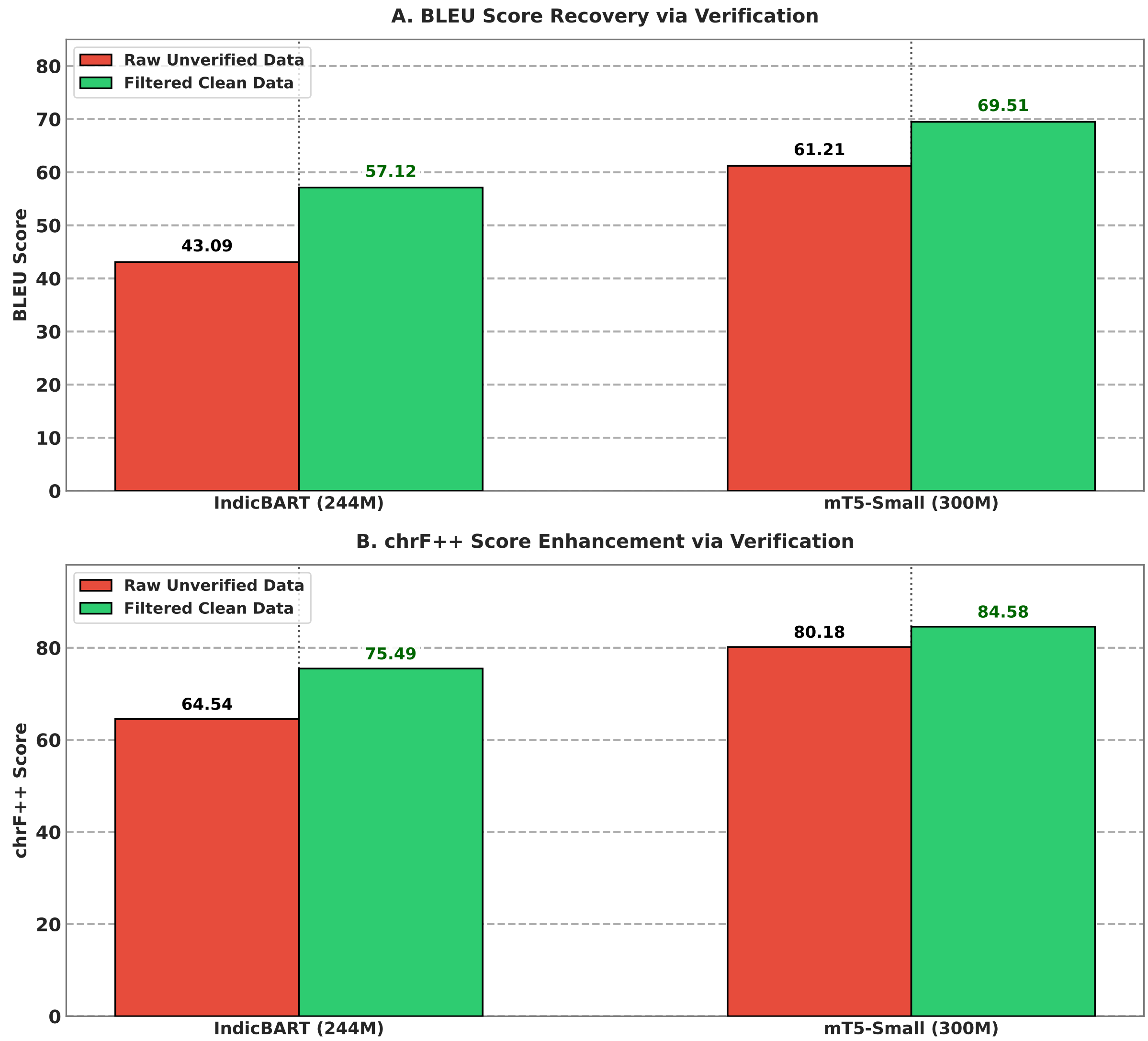

Figure 8: Effect of verification filtering on BLEU (A) and chrF++ (B).

Training on unverified data led to a severe drop in performance (−14.03 BLEU for IndicBART, −8.30 BLEU for mT5-small). Unfiltered LLM outputs had hallucinated sentences which corrupted the gradients during training. Multi-tier verification removes the flawed pairs successfully recovering the BLEU scores to 57.12 (IndicBART) and 69.51 (mT5-small).

IndicBART shows sensitivity to noise and corrupted data, with a 24.6% relative BLEU drop. mT5-small showed a smaller relative drop (11.9%). This could be directly attributed to mT5's massive pre-training corpus. This highlights the importance of the quality audit of data while using LLMs for generating or correcting the data.

5 Conclusion

This study builds and presents a reproducible framework for text normalization across three Marathi dialects. By the efficient use of LLMs, it solves the problem of not having enough parallel data for low-resource Indic language varieties instead of relying on manual annotations. Also, the verification engine proves to be effective since its correction ability prevented the models from performance degradation.

Our experiments show that mT5-small reaches 69.51 BLEU and 84.58 chrF++ on the expanded multi-dialect benchmark, beating IndicBART by a score of 12.39 BLEU. Also, our multi-tier verification step stopped synthetic data quality from collapsing as it recovered +14.03 BLEU.

These results offer a decent starting point for normalizing other Indic sub-dialects, but a few limitations remain. Firstly, LLMs add up to the extra cost of infrastructure and compute. Also, synthetic expansion comes at a cost of degraded performance for dialects that already perform well (Varhadi dialect in this case). Future work should test the performance around speech-based systems, not just text-based metrics. Overall, our verified pipeline offers a scalable way to expand digital language support for low-resource regional dialects.

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